

HARLEEN BUTTAR

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EDUCATION

M.S. Applied Data Science, University of Chicago | GPA: 3.78

Expected August 2026

Coursework: Machine Learning, Big Data & Cloud Computing, Time Series Analysis, Algorithmic Marketing, Statistical Models, Data Engineering, NLP

B.S. Data Science, University of Chicago

June 2025

DATA SCIENCE EXPERIENCE

Data Science Project Manager

June 2025 - September 2025

CharacTour, Chicago, IL

- Directed 3 data science interns analyzing 10M+ consumer records; built Python NLP pipelines from TikTok transcripts detecting cluster-specific speech patterns across sentence structure and emotional cadence to power identity-driven AI recommendations for 23 personality segments
- Developed scalable automated diarization pipeline to slice, transcribe, and assign speaker identities across long-form video scenes, reducing manual tagging by 80%

Team Member, Data Science Clinic

January 2025 - June 2025

Internet Equity Initiative, University of Chicago, Chicago, IL

- Built and deployed interactive broadband equity dashboard in Plotly and Plotly Dash to visualize disparities in internet access across geographies and demographics
- Partnered with policy researchers to translate continuous data from thousands of devices into actionable insights informing broadband infrastructure and digital equity recommendations

GRADUATE PROJECTS

QLoRA Recipe Chef (Machine Learning 2)

March - June 2026

- Developed an end-to-end recipe generation system by fine-tuning the Qwen2.5-1.5B-Instruct model with QLoRA, 4-bit quantization, and LoRA adapters on the RecipeNLG dataset, enabling controllable recipe generation conditioned on cuisine and dietary preferences while training only 0.6% of model parameters.
- Built a complete ML evaluation and deployment pipeline, including custom cuisine/diet labeling, automated metrics (ROUGE, BERTScore, perplexity, ingredient fidelity, and knowledge retention), and an interactive Gradio application; optimized Google Drive-backed caching to reduce experiment rerun time from ~90 minutes to ~5 minutes.

Clothing Fit Prediction & Size Recommendation (RentTheRunway / ModCloth)

January - March 2026

- Built 3-phase ML pipeline across 192,000+ transactions: unsupervised segmentation (K-Means, GMM, DBSCAN, K-Modes, PCA, LDA), 7-classifier fit prediction (SVM, Random Forest, CART, KNN, Naive Bayes) with SMOTE and stratified cross-validation, and collaborative filtering recommender
- Benchmarked all classifiers on multiclass fit prediction (Small/Fit/Large) with per-class precision, recall, and F1; tested cross-retailer generalization by training on ModCloth and evaluating on RentTheRunway

UK Fruit and Vegetable Price Forecasting (Time Series Analysis)

January - March 2026

- Engineered 184-feature weekly dataset incorporating agricultural indices, UK fuel prices, and freight indices with energy crisis shock indicators; benchmarked ARIMA, Holt-Winters, VAR, LM-ARIMA, SARIMA, ARIMAX, DeepAR, and Mamba across 20 commodities with Granger causality and stationarity testing on a 2025 holdout set

Large-Scale Sentiment-Rating Mismatch Detection (Big Data)

September - December 2025

- Built distributed PySpark NLP pipeline on Google Cloud Dataproc applying TF-IDF, Word2Vec, K-Means, LSH, and LDA at scale on Amazon review data; managed schema validation, data type compatibility, and fault tolerance with outputs written to GCS

RELEVANT EXPERIENCE

Sourcing and Due Diligence Intern — Angeles Investors, Chicago, IL

May - August 2024

- Evaluated 50+ startups across business models, market positioning, and competitive risk; conducted extensive due diligence identifying key risks, opportunities, and trends to inform investment decisions

Data Analyst Intern — Prospanica, Chicago, IL

June - August 2024

- Analyzed organizational datasets with SQL and Excel to uncover trends; presented data-driven insights to leadership supporting geographic strategic planning and educational initiative decisions

TEACHING EXPERIENCE

Teaching Assistant, Data Engineering & Data Science Clinic — University of Chicago

September 2025 - June 2026

- Support graduate students through pipeline design, SQL, Python, cloud infrastructure, and GitHub; guide code quality standards, collaborative development practices, and technical communication across demanding coursework

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, scikit-learn, PySpark, Plotly, Selenium, NLTK, Gensim, VADER), R, SQL (Postgres, MySQL, SQL Server), DuckDB, Java

ML & Analytics: Supervised/Unsupervised ML, NLP, Causal Inference, Time Series Forecasting, Recommendation Systems, Feature Engineering, Statistical Modeling

Tools & Platforms: Tableau, Plotly Dash, GitHub, GCP (Dataproc, GCS, GKE), AWS, Databricks (familiar)

Languages: English, Punjabi, Spanish (all fluent)